Highlights

Forgeable greenbeards: bistability and hollow collapse in the evolutionary dynamics of certified agent populations

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- A forgeable identity tag turns certification into a nonlinear population flow
- In-group conduct is a security parameter: forgery tolerance 0.866 versus 0.400
- The protective fine diverges as forgery becomes undetectable
- Behavioural mimics hollow out the tag before conduct ever degrades
- The flow is bistable, so certification buys robustness and not a safer level

Forgeable greenbeards: bistability and hollow collapse in the evolutionary dynamics of certified agent populations

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ARTICLE INFO

Keywords:
evolutionary game theory
replicator dynamics
bistability
tag-based cooperation
greenbeard
certification
multi-agent systems

ABSTRACT

A greenbeard is a signal that lets its bearers treat one another differently from everyone else, with the classical vulnerability that impostors can wear it. Machine agents now carry such signals: provenance attestations and certification marks, cheap to check and forgeable. We make the mark a strategic variable in a four-strategy model of a competitive development race, holding the game fixed so that every effect belongs to the identity layer, and analyse the resulting 48-strategy replicator flow and its finite-population counterpart. A design is a triple: a badge that is genuine, forged or absent, plus the conduct it runs towards partners who pass verification and towards everyone else. A forged badge passes with probability σ. Every transition we report is a transversal eigenvalue crossing zero, so each critical value is exact. Identity earns its keep only below a reciprocity threshold, where deterrence fails. A club that merely reciprocates within its membership tolerates more than twice the forgery of one that trusts unconditionally, so in-group conduct is a security parameter. The protective fine diverges as forgery becomes undetectable, so penalties cannot substitute for detection. Collapse is graceful in the wrong way: a behavioural mimic invades before the exploiter, hollowing out the credential while conduct still looks sound, and better interaction structure leaves it the only failure mode. The flow is bistable with no badge-mixed attractor, so certification buys robustness rather than a lower level of harm, and what holds it up is the rent it charges outsiders.

1. Introduction

Machine agents are beginning to carry identity. Agent-to-agent protocols publish identity and capability metadata before agents exchange work [1]; regulatory technical-documentation duties require providers to identify an AI system or general-purpose model and its version [2]; and proposals for agent attestation infrastructure are being drafted while the systems they would govern are being deployed [3–5]. Even where nothing is formally attested, model evaluators can recognise and favour their own generations, while language models in general favour machine-generated communications [6, 7].

An identity signal that changes how its bearers treat one another is a *greenbeard*, and evolutionary biology has known for fifty years both what greenbeards can do and how they fail [8, 9]. They can sustain cooperation among strangers with no repeated interaction and no kinship, because the tag substitutes for both [10, 11]. They fail when the tag comes apart from the behaviour it advertises, so that an impostor wears the beard and takes the benefit without paying the cost [12–14]. The biological literature is largely concerned with tags that are hard to fake, because a gene cannot easily choose its own phenotype. An agent's badge is not like that. Whether a mark is forgeable is a property of the verification technology [15, 16], and therefore a design choice, and therefore something a governance regime can spend money on.

The question this paper answers is what a population of agent operators does with that choice. The question is not rhetorical, because two forces pull against each other. Certification and third-party quality disclosure are established mechanisms for addressing information problems [17]. Here the certification mark is also a private cost: it must be obtained, maintained and audited, and by assumption every unit of that cost burdens the compliant operator and never the impostor. A certification mark is also a private benefit: it buys access to partners who will behave well towards whoever carries it. Which force wins is not a matter of taste, and neither is the policy question that follows, namely

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whether an authority does better to spend on harder attestation, on penalties for forgery, on cheaper compliance, or on the interaction structure that decides who meets whom.

We answer both by making the badge a strategic variable in an evolutionary model. The interaction layer is the reduced four-strategy AI race of Fernández Domingos and Han [18], used without modification, so that every effect we report is attributable to the identity layer we build on top of it. A design is a triple $(b, s_{\text{in}}, s_{\text{out}})$: a badge that is genuine (G), forged (F) or absent (N), the reduced race design run against a partner whose badge passes the focal agent's check, and the design run against everyone else. Operators do not optimise; designs that pay better are copied more, in the standard replicator and finite-population dynamics of evolutionary game theory [19–21].

Our contribution is an explanation, not a method or a benchmark. In one sentence: *a forgeable identity mark buys an agent ecosystem robustness rather than safety, and pays for it with an entry barrier that cannot be legislated away, because the exclusionary conduct that creates the barrier is exactly what makes the mark survive forgery.* The results below make that precise.

The model as a dynamical system. The object we analyse is a nonlinear flow. Forty-eight designs compete on a 47-dimensional simplex under replicator dynamics, and the same payoff structure drives a finite-population birth-death process. Every result below is a statement about that flow, and it takes one of two forms. Most are critical parameter values at which a transversal eigenvalue at a vertex of the simplex changes sign, so that a rest point exchanges stability with an invading direction. Because the underlying race is evaluated exactly rather than sampled, and because each of those eigenvalues is affine in the parameters, every such critical value is a simple root available in closed form; numerical integration confirms them rather than producing them. The remaining results concern the global structure of the flow rather than local stability at a vertex: the flow turns out to be bistable, with two attractors lying on disjoint faces of the simplex, and it is the two basins, not the equilibrium level of harm, that carry the eighth result. Nonlinearity is not incidental here. It is why a regime whose credential works and one whose credential means nothing can be indistinguishable from the outside, and why an ecosystem that has fallen out of the certified regime does not climb back into it.

Results. We show, first, that identity is the instrument of last resort rather than of first choice. The race has a reciprocity threshold $L_R = 0.551$ (Proposition 1): below it, and below an assortment $r^\dagger = 0.076$, a first-striker profitably exploits any safe unbadged resident, because the half-step lead it steals in the opening round converts into the prize and reciprocity punishes it only in kind. Where liability is enforceable, so that L lies above L_R, the identity layer is close to worthless, and the certification value we measure falls from an average of 0.054 below L_R to 0.0025 above it. Identity earns its keep precisely in the regime where harm cannot be traced to the seat that caused it [3, 22].

Second, in-group conduct is a defensive parameter, not an ethical one. In a well-mixed population a club that reciprocates inside its membership tolerates $\sigma^* = 0.866$ spoof success before an exploiter invades; one that trusts verified partners unconditionally tolerates only 0.400 (Proposition 2). At the baseline assortment the pair is 0.962 against 0.444. The ratio is 2.17 either way. A club is safest when its members are conditional cooperators, not when they are saints, because unconditional trust is what an impostor is buying access to.

Third, fines cannot buy what detection does not deliver. Setting the invasion condition to zero and solving for the penalty gives a protective fine that diverges as $\sigma \to 1$ (Proposition 3): well mixed, 11.9 at $\sigma = 0.90$, 58.2 at 0.95, 428.6 at 0.99. This divergence is the Becker trade-off between penalty size and detection probability [23]. Its novel feature here is that in an agent economy the detection probability is fixed by cryptography rather than by how many inspectors are hired, so the usual policy substitution is unavailable.

Fourth, verification quality is irrelevant to whether a club can start. Against any resident that does not condition on badges, a genuine badge earns exactly its unbadged twin's payoff minus the dues, at every σ (Proposition 4). Clubs therefore nucleate only above an assortment threshold, $r^* = 0.063$ at our baseline (Proposition 5), and no amount of attestation engineering substitutes for it.

Fifth, collapse degrades in the wrong order. The behavioural mimic, a forger whose conduct is identical to the club's, invades at $\sigma_m = 0.938$, below the exploiter's threshold (Proposition 6). Between the two thresholds the ecosystem still behaves well while the mark has stopped meaning anything: at $\sigma = 0.97$ forgers hold 55% of the population, the posterior probability that a passed badge is genuine has fallen to 0.05, and the unsafe frequency is 0.058 against the baseline 0.090. An observer monitoring conduct would see nothing wrong, and would in fact record an improvement.

Which of the two invaders arrives first is decided by the interaction structure, and in a direction that undercuts the instrument of the fourth result. Clustering penalises an exploiter, which meets its own aggression in a self-encounter, and not a mimic, which meets the club's own conduct. The exploiter's threshold therefore rises from 0.866 to 1 by $r = 0.134$ while the mimic's moves by 0.001, and the two cross at $r = 0.077$. Above $r = 0.134$ the only way into a club is impersonation by an agent that behaves, so improving the interaction structure removes the visible failure mode and concentrates all remaining risk in the invisible one.

Sixth, the collapse is not reversible. At zero assortment a club cannot re-enter a population that has collapsed to forgery or to anarchy, at any verification quality whatsoever. Keeping a mark is strictly easier than rebuilding one.

Seventh, scoping marks to providers makes concentration a safety variable. With K symmetric provider clubs whose marks do not recognise each other, the ecosystem unsafe frequency is exactly $1 - \text{HHI}$ (Proposition 7): a duopoly runs at 0.500 and eight providers at 0.875, while federated recognition is safe at any concentration. The same fragmentation dissipates the exclusion rent, from 30.4 under a single club to 2.5 across eight, which is why the incentive to federate is weakest exactly where federation would help most [24].

Eighth, and the result that reframes the other seven: what certification buys a settled market is not safety but an entry barrier. The replicator flow is bistable and has no badge-mixed attractor (Proposition 8). All 200 interior starts reach a regime that is either entirely badged, 44% of them, or entirely unbadged, and both are internally safe, with the largest unsafe frequency across all attractors below 10^{-21}. Since both are safe, the certified regime's social payoff is lower than the uncertified one by exactly the dues, 57.0 against 59.0 (Corollary 9). The small-mutation result is not contradicted: certification does lower harm there, from 0.159 to 0.090, but what it lowers is the harm of a market being knocked out of safety, not of one sitting in it. Certification buys robustness, not level.

The exclusion is paid instead at the boundary, as a toll. A compliant unbadged outsider that plays the club's own conduct towards everyone earns 26.64 where a member nets 57.00, a penalty of 30.36 for no conduct-related reason, and the encounter runs unsafe at 0.461 (Proposition 10). A club that charges no such toll charges -2.00, is behaviourally indistinguishable from an unbadged cooperator, has spoof threshold exactly zero and never survives. Across the four out-group policies the toll and the forgery tolerance order identically, with nothing in between. Fining forgery restores the gentle club's threshold to 0.923 but not its share, which stays below 0.032, because a fine changes what an impostor faces and not what a badge earns against agents who never check one (Corollary 11). The exclusion is not an unfortunate side effect of certification that better rules could remove. It is the load-bearing element.

Why this matters beyond the model. Agent identity standards are being written now [1, 25, 26], and the case for them is usually made in terms of accountability [27]: if we know which agent did what, we can hold someone responsible. Our results say that identity delivers something different from accountability. It delivers a club, and clubs are safe inside and hostile at the edge. The policy question is therefore not whether to build attestation but what to require of the conduct it gates, and our answer is uncomfortable: the requirement that would remove the harm also removes the club. Managed mutual recognition and equitable interoperability are established institutional proposals [28, 29]. In our model, such federation converts the boundary into an interior without asking any club to be gentle to strangers.

1.1. Related work

Evolutionary dynamics as a nonlinear system. The formal setting is evolutionary game dynamics [19, 20, 30]: a population of competing designs is a nonlinear flow on a simplex, qualitative changes of behaviour are bifurcations of that flow, and the objects of interest are collective rather than individual [31, 32]. That programme has established that transitions between cooperative and defecting phases are sharp and often reducible to a universal form [33, 34], and that much of the outcome is decided by who meets whom, through networks, layers or conformity [35–38]. Tags have been studied there as generators of assortment without memory, as phenotypic similarity in well-mixed populations [39] and as a mutating tag space in finite ones [40]. Two features separate this system from that body of work. First, the tag is forgeable at a rate the modeller sets, so the bifurcation parameter is a property of a verification technology rather than of the players, and the comparative statics are with respect to an engineering budget. Second, the flow carries two attractors differing in whether the tag exists at all rather than in how much cooperation it supports, so what matters is basin structure and not equilibrium level. Whether either regime can arise at all turns on a classical condition: altruism invades only under positive assortment [41, 42], supplied here by one provider's agents sharing a platform. By Proposition 4 the badge contributes nothing to nucleation, so assortment is not one route among several but the only one.

Greenbeards, forgeable tags and parochial conduct. Kin selection [8, 43] explains why a gene favours copies of itself in others, and a green beard is a gene carrying both marker and behaviour [9]. Riolo et al. [10] showed that tag similarity alone sustains cooperation without reciprocity, and Roberts and Sherratt [12] that this turns on cost, since a cheap tag is cheap to fake, which the original authors disputed [44]. Jansen and van Baalen [13] showed that a continuously mutating tag space outruns impostors, and Gardner and West [14] reviewed the empirical cases, which are real but rare [45, 46]. In biology, forgeability is a fact about the organism; in an agent economy it is a fact about the verification infrastructure [15, 16], so σ is a policy variable. The closest structural ancestor is Hammond and Axelrod [47], whose agents carry a tag and two conditional strategies, our design triple minus the badge type. Their tags are inherited faithfully and cannot be faked, so our question does not arise: their result explains why differential treatment emerges, ours why it cannot be removed from a certification regime without removing the regime. Choi and Bowles [48] pair in-group generosity with out-group hostility too, but need group selection and explicit intergroup warfare, while Masuda and Fu [49] give the standard condition that tag and strategy be inherited as one unit, satisfied here by construction. We need neither: one well-mixed pool with assortment, and nothing won by fighting. Out-group harshness is favoured here because it is the only thing that makes a costly membership signal worth acquiring honestly, so excludability, not conflict, carries the result.

Signalling, certification and enforcement. In the handicap tradition signals are honest because they are differentially costly to fake [50, 51], and market signalling separates types for the same reason [52]. Our forger pays less than the compliant member, not more. Signal honesty need not rest on a realised equilibrium cost [53]; in our model it rests instead on detection. The setting is nearer a market degraded by failed quality signalling [54] and certification intermediaries [17, 55]; the attribute our badge certifies is never learned even after the encounter, which makes the badge a credence rather than an experience good [56, 57]. Penalty and detection are classically substitutes [23]; Proposition 3 states the limits of that substitution when detection is technological rather than budgetary.

AI races and agent identity. The interaction layer is the reduced race of Fernández Domingos and Han [18], descended from the development race of Armstrong et al. [58] and formalised evolutionarily by Han et al. [59], whose programme covers voluntary commitments, heterogeneity, auditing and trust [60–63]. Those instruments act on the race's payoffs or on an external monitor; ours acts on who is recognised as an insider, which Proposition 1 shows is the only lever available in the regime where the payoff levers have no purchase. Applied work on agent governance treats visibility and provenance as prerequisites [3–5, 22], and Hammond et al. [64] name selection pressure and mis-identification among multi-agent risks without formalising either. Work on model-origin and counterpart cues documents self-favouring evaluation and identity-conditioned interaction [6, 65, 66]. Model outputs carry detectable idiosyncratic signatures [67], but models show no general, consistent self-recognition [68]. We therefore treat recognition as available but imperfect rather than free and reliable. What that strand lacks, and we supply, is the population-level consequence: what an ecosystem converges to when recognition is available, imperfect and worth faking.

2. Model

2.1. The interaction layer

The interaction layer is the reduced race of Fernández Domingos and Han [18], reproduced without modification. Two seats repeatedly choose Safe or Unsafe. Unsafe pays more in the stage game and advances the race faster, but accumulates private setback risk for whoever leads at the horizon. The horizon is stochastic: $T = 5 + G$ with G geometric at rate 0.2, so $\mathbb{E}[T] = 9$. Four reduced designs are considered, in increasing order of the harm they cause: AS is always safe, CS opens safe and mirrors, CAS opens unsafe and mirrors, and AU is always unsafe.

Because the four designs are deterministic, the action path of an ordered pair is fixed once the pair is fixed, and every matchup is evaluated by exact expectation over the horizon law rather than by sampling. We write A for the 4×4 matrix of expected race payoffs, M for the expected number of Unsafe actions of the focal seat, and U for the expected fraction of Unsafe rounds; Table 1 gives A and M.

2.2. The identity layer

Each seat is operated by an agent that carries a badge and conditions its conduct on the badge its partner presents [65, 66]. A design is a triple $(b, s_{\text{in}}, s_{\text{out}})$ with $b \in \{\mathsf{G}, \mathsf{F}, \mathsf{N}\}$ and $s_{\text{in}}, s_{\text{out}}$ reduced race designs, so the design space has $3 \times 4 \times 4 = 48$ elements.

A badge either passes a verification check or does not. Missing mandatory validation or attestation is a documented security risk in agent protocols [16]. We represent that risk by letting a genuine badge always pass, an absent badge never pass, and a forged badge pass with probability σ, the *spoof success probability*; $1 - \sigma$ is the detection power of the attestation technology. Verification happens once, at the start of a race, and holds for its duration. That matches the interaction layer, in which the executed design of a seat is fixed before the first round. The two checks of an encounter concern two different badges, so they are independent by construction, and every pairwise expectation is an exact four-term sum,

$$a(i,j) \;=\; \sum_{\substack{x\in\{\text{in},\text{out}\}\\ y\in\{\text{in},\text{out}\}}} w_x(q_j)\, w_y(q_i)\; A\big(s^i_x,\, s^j_y\big), \qquad w_{\text{in}}(q) = q, \quad w_{\text{out}}(q) = 1-q, \tag{1}$$

where q_i is the rate at which design i's badge elicits the in-group response. The harm matrix $m(i,j)$ and the behaviour matrix $u(i,j)$ are the same sum with M and U in place of A.

Carrying a badge costs something. A genuine badge costs κ_g per race, the price of issuance, attestation and compliance; a forged badge costs κ_f, and the interesting regime is $\kappa_f < \kappa_g$. A detected forgery may additionally be fined ρ, so a forger expects to pay $(1-\sigma)\rho$ per encounter.

Detection has two consequences that the model keeps separate, because they are different instruments. Under *screening* the checking agent responds to a failed badge by playing s_{out}: verification happens in real time and the counterparty can act on it. Under *retrospective auditing* the fine is levied but the behavioural response is not available, because the detection arrives after the interaction; formally, $q_{\mathsf{F}} = 1$ while the expected fine is still $(1-\sigma)\rho$. Section 3.6 separates the two.

Finally, agents do not meet uniformly at random. With probability r an agent interacts within its own provider cluster, modelled as meeting its own design [69], and with probability $1-r$ it meets a uniform draw from the population. This is the one parameter the identity layer does not create: it is the pre-existing structure of the agent economy, in which agents of one provider co-occur on one platform.

2.3. Two functionals

The *private* functional is what the operator of a seat receives,

$$\pi_P(i,j) \;=\; a(i,j) \;-\; L\, m(i,j) \;-\; c(b_i) \;-\; f(b_i), \tag{2}$$

with L the liability charged per Unsafe action of one's own seat, c the badge cost and f the expected fine. The *social* functional counts the full harm and the real resource costs, but not fines, which are transfers:

$$\pi_S(i,j) \;=\; a(i,j) \;-\; h\, m(i,j) \;-\; c(b_i). \tag{3}$$

Only π_P drives the dynamics; π_S is the yardstick. The baseline sets $L = 0$. That setting is the regime the paper is about: where harm cannot be attributed to the seat that caused it, the liability channel is closed [3, 22] and identity is the only channel left. Section 3.9 reopens it and measures what identity is worth as a function of L.

2.4. Dynamics, observables and parameters

We report two dynamics because they answer two different questions about the same game. The finite-population pairwise-comparison process in the small-mutation limit [21, 70] answers which single design the market spends its time in, since under rare mutation the population is monomorphic almost always. The replicator flow [19, 20], run from many interior initial conditions, answers a different question: from an arbitrary starting composition, which regime does the market settle into, and how large is each basin. Both act on the assortment-adjusted private functional. The replicator flow is integrated with EGTtools [71]; the small-mutation chain uses a closed-form birth-death fixation probability, which is what makes the 48-design space tractable, and is checked against the full EGTtools chain on reduced spaces.

The flow. Write Δ for the simplex of population states over the 48 designs and $x \in \Delta$ for a state. With assortment r, the expected private payoff of design i is

$$f_i(x) \;=\; r\,\pi_P(i,i) \;+\; (1-r)\sum_j x_j\, \pi_P(i,j), \tag{4}$$

and the replicator flow is

$$\dot{x}_i \;=\; x_i\big[\, f_i(x) - \bar{f}(x)\big], \qquad \bar{f}(x) \;=\; \sum_k x_k f_k(x). \tag{5}$$

Table 1
The interaction layer, exactly evaluated over the horizon law: expected race payoff A (left) and expected number of unsafe actions M (right) of the row design against the column design. The row that drives every result is CAS against CS: a first strike earns 61.63 where the resident earns 59.00 against itself, at the cost of 4.78 unsafe actions, so reciprocity deters it only above $L_R = 0.551$.

	payoff A				unsafe actions M			
	AS	CS	CAS	AU	AS	CS	CAS	AU
AS	59.00	59.00	8.60	5.40	0.00	0.00	0.00	0.00
CS	59.00	59.00	26.64	16.60	0.00	0.00	4.22	8.00
CAS	101.77	61.63	27.20	27.20	1.00	4.78	9.00	9.00
AU	48.64	47.36	27.20	27.20	9.00	9.00	9.00	9.00

Every vertex e_j of Δ is a rest point of Equation (5) at every parameter value, and the eigenvalue of the linearisation at e_j transversal to that vertex in the direction of design i is

$$\lambda_{i|j} \;=\; f_i(e_j) - f_j(e_j) \;=\; r\,\pi_P(i,i) \;+\; (1-r)\,\pi_P(i,j) \;-\; \pi_P(j,j). \tag{6}$$

Each $\lambda_{i|j}$ is affine in the spoof success σ, in the fine ρ, in the badge costs κ_g and κ_f and in the liability L, because the handshake of Equation (1) is affine in the pass rates and the functional of Equation (2) is affine in the rest. Every threshold reported in Section 3 is therefore the unique simple root of one such eigenvalue in one parameter. The qualitative event each threshold marks is a transcritical bifurcation of the flow restricted to the edge $\{e_i, e_j\}$, on which Equation (5) reduces to $\dot{x} = x(1-x)\,[\,f_i - f_j\,]$ with the bracket affine in x: as the parameter crosses the threshold, the interior edge equilibrium passes through the vertex and exchanges stability with it. This is why the thresholds below are exact rather than estimated, and it is what the closed forms in Section 3 are expressions for.

The distinction between the two is not a robustness check, and it needs one piece of care that is easy to skip. The replicator flow of this game turns out to be multistable, with attractors on disjoint faces of the simplex (Proposition 8). Every observable here is bilinear in the population state, so evaluating one at the arithmetic mean of two disjoint attractors invents encounters between designs that never actually meet. We therefore evaluate every observable *at each attractor* and average the results over basins, never the reverse. Section 3.8 reports what the two conventions differ by, because the gap is large enough to change the sign of a headline effect.

Beyond the population unsafe frequency U and the social payoff, we report three observables specific to the identity layer. The *attestation integrity* is the posterior Pr(badge genuine | badge passed), the epistemic content of a passed check. The *mark lift* is Pr(partner behaves safely | its badge passed) − Pr(partner behaves safely | it did not), the behavioural value of a passed check to the agent that performed it. The *harm decomposition* assigns every ordered encounter to one of four blocks by the badge status of the two seats and splits U across them exactly.

Baseline parameters are $\sigma = 0.5$, $\kappa_g = 2$ (about 3.5% of the 57.0 a safe club member nets), $\kappa_f = 0$ (forgery is free, the worst case), $\rho = 0$ (fines are an instrument, not part of the world), $r = 0.1$, $L = 0$, $h = 20$, population size $Z = 100$ and selection intensity $\beta = 0.05$. Every one of them is swept.

Table 2
Closed-form thresholds of the identity layer at zero liability. σ^* is the spoof success above which the unconditional forger invades the certified reciprocator club, σ_m the threshold for the behavioural mimic, and r^* the assortment below which no club nucleates, whatever the verification. Dues weaken the club; fines strengthen it only through the detections the screen already makes.

dues κ_g	fine ρ	σ^* (forger)	σ_m (mimic)	r^*
0.5	0	0.909	0.985	0.016
0.5	5	0.921	0.987	0.016
0.5	20	0.942	0.990	0.016
1.0	0	0.895	0.969	0.031
1.0	5	0.908	0.973	0.031
1.0	20	0.933	0.981	0.031
2.0	0	0.866	0.938	0.063
2.0	5	0.883	0.946	0.063
2.0	20	0.915	0.962	0.063
4.0	0	0.807	0.876	0.126
4.0	5	0.832	0.893	0.126
4.0	20	0.878	0.924	0.126
8.0	0	0.691	0.753	0.252
8.0	5	0.730	0.786	0.252
8.0	20	0.805	0.847	0.252

Table 3
The baseline equilibrium across nested design pools, in the small-mutation limit ($\sigma = 0.5$, $r = 0.1$, $\kappa_g = 2$, $L = 0$, $Z = 100$, $\beta = 0.05$). The identity layer roughly halves the long-run unsafe frequency of the uncertified world; forgery at a coin-flip spoof rate claws back about a quarter of the gain that unforgeable badges would deliver.

design pool	unsafe	club	falsebeard	integrity	social
certified world (48 designs)	0.090	0.42	0.07	0.92	39.1
unforgeable badges (32)	0.068	0.47	0.00	1.00	43.6
uncertified world (16)	0.159	0.00	0.00	1.00	25.4
raw race (4)	0.159	0.00	0.00	1.00	25.4

Table 4
The symmetric K-provider world under provider-scoped marks. Cross-provider checks fail, so the ecosystem unsafe frequency is $1 - \mathrm{HHI}$ exactly; federated recognition plays the club design throughout and is safe at any concentration. The exclusion rent is the payoff gap between a member and a compliant unbadged entrant.

K	HHI	unsafe (scoped)	unsafe (federated)	member	rent
1	1.000	0.000	0.000	57.0	30.4
2	0.500	0.500	0.000	41.1	14.5
3	0.333	0.667	0.000	35.8	9.2
4	0.250	0.750	0.000	33.2	6.5
8	0.125	0.875	0.000	29.2	2.5
20	0.050	0.950	0.000	26.8	0.1

3. Results

Table 5 is a map of what follows. Each row is one qualitative change in the flow of Equation (5), together with the parameter that drives it, the closed form that locates it and its value at the baseline. The subsections then take them in order and say what each one means for a certification regime.

Table 5
The qualitative transitions of the identity layer. Rows one to four and row six are transcritical bifurcations of Equation (5) restricted to an edge of the simplex: a transversal eigenvalue $\lambda_{i|j}$ of Equation (6) passes through zero and the resident vertex exchanges stability with the invading direction. Row five is a codimension-two degeneracy at which two such eigenvalues vanish together, so that the ecosystem changes which of its two failure modes it meets first. Row seven is not a bifurcation of the flow but a divergence of the locus of one: the curve in the (σ, ρ) plane along which the club is marginally protected has a vertical asymptote at $\sigma = 1$. Row eight is a global property that no local calculation reveals.

transition	parameter	location	value
safe unbadged resident loses stability to a first striker	liability L	Eq. (7)	$L_R = 0.551$
certified club loses stability to an unconditional forger	spoof success σ	Eq. (9)	$\sigma^* = 0.866$ at $r = 0$, 0.962 at $r = 0.1$
certified club loses stability to a behavioural mimic	spoof success σ	Eq. (13)	$\sigma_m = 0.938$
unbadged resident loses stability to a certified club	assortment r	Eq. (12)	$r^* = 0.063$
the two invading eigenvalues vanish together and the invaders exchange order	(r, σ) jointly	Eq. (9), (13)	$r = 0.077$
the forger's threshold leaves the admissible range of σ	assortment r	Eq. (9)	$r = 0.134$
the locus of marginal protection escapes to infinity	(σ, ρ) jointly	Eq. (10)	pole at $\sigma \to 1$
two attractors on disjoint faces of the simplex	none	Prop. 8	basins 0.44, 0.56

3.1. Why identity is needed: the reciprocity threshold

The race has a property that decides what any identity layer has to do. Consider a population of unbadged reciprocators, all playing CS: open safe, then mirror. A rare CAS mutant opens unsafe, takes a half-step lead in the first round, and is mirrored from the second round onwards. Reciprocity punishes it, but only in kind, and the stolen half-step survives to the horizon and converts into the prize. The mutant earns 61.63 against the resident's 59.00, at the cost of 4.78 extra unsafe actions.

Proposition 1 (Reciprocity threshold)**.** *The advantage of the first-striker is affine in the liability and vanishes at*

$$L_R = \frac{A(CAS, CS) - A(CS, CS)}{M(CAS, CS) - M(CS, CS)} = 0.551. \tag{7}$$

For every $L < L_R$ and every assortment $r < r^\dagger$, the unbadged first-striker (N, *CAS*, *CAS*) *invades every unbadged resident that plays safe conduct in self-play, where*

$$r^\dagger = \frac{A(CAS, CS) - A(CS, CS)}{A(CAS, CS) - A(CAS, CAS)} = 0.076. \tag{8}$$

In that region no unbadged population sustains safe play, whatever its conduct rule.

Both bounds are needed, and the second is as informative as the first. The first-striker's transversal eigenvalue at a safe unbadged resident is $A(\text{CAS}, \text{CS}) - A(\text{CS}, \text{CS}) - r\,[A(\text{CAS}, \text{CS}) - A(\text{CAS}, \text{CAS})] = 2.631 - 34.431\,r$: clustering makes a first-striker meet its own aggression, and by $r^\dagger = 0.076$ that self-encounter has eaten the whole stolen half-step. Above $r^\dagger$ the four residents (N, $\cdot$, CS) resist it, which is why the uncertified regime of Proposition 8 is a safe one even at $L = 0$. The proposition is checked exhaustively over the eight safe unbadged residents rather than argued. Its content is that within-race reciprocity is not enough on its own: the race rewards the opening move, a mirror cannot undo an opening move, and what rescues a safe unbadged population is the very assortment that Section 3.4 shows a club also needs. Something outside the race must distinguish partners before the first round, and that is what a badge does.

Figure 1B shows the consequence. At $L = 0$ and no assortment, the uncertified world runs at unsafe frequency 1.000: it is complete anarchy. The value of identity is therefore not a marginal improvement on a functioning market.

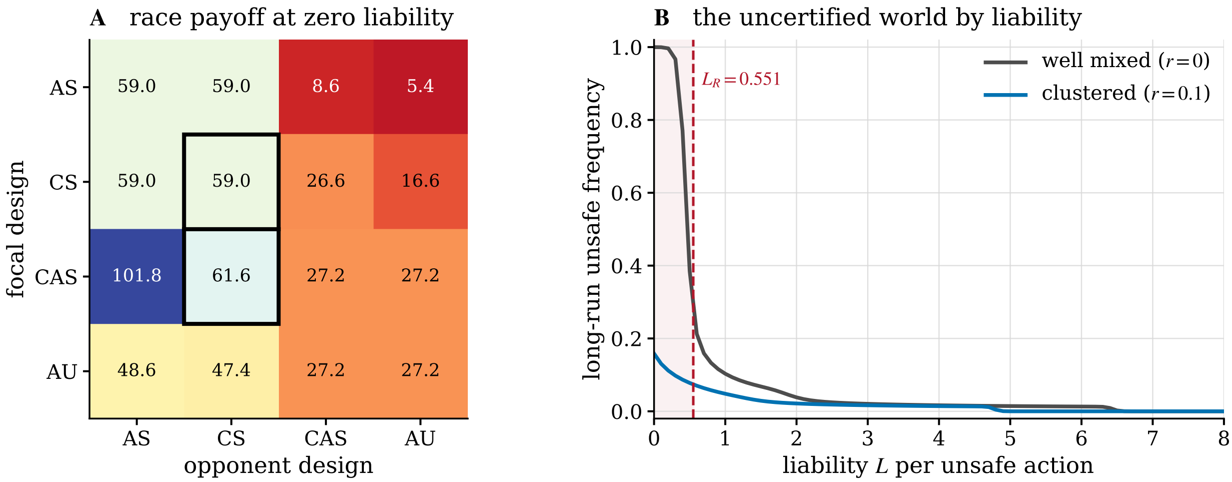

Figure 1: The race rewards the opening move, so reciprocity alone cannot hold a safe population. (A) Private race payoff $A - LM$ at $L = 0$. The boxed cells are the handshake exploit: a CAS first-striker earns 61.63 against a CS resident that earns 59.00 against itself. **(B)** Long-run unsafe frequency of the uncertified world (badge N only) as liability rises. The shaded region is $L < L_R = 0.551$, where Proposition 1 rules out any safe unbadged population in a well-mixed one. Without clustering the world is fully unsafe there; the clustered baseline sits above $r^{\dagger} = 0.076$, so some safe unbadged residents survive, and it is better but still far from safe.

It is the difference between a market and no market. Conversely, once L rises above L_R the liability channel does the work by itself, and Section 3.9 shows the identity layer becoming worthless there.

3.2. How much forgery a club survives

A certified club is a design (G, u, v): carry a genuine badge, play u towards partners whose badge passes, v towards everyone else. A forger $(\mathsf{F}, w, \cdot)$ buys access to the in-group conduct u with probability σ and pays κ_f rather than κ_g.

Proposition 2 (Spoof threshold)**.** *For an unconditional forger* (F, w, w) *against a club* (G, u, v) *in a well-mixed population, the invasion condition is affine in* σ *and the threshold is*

$$\sigma^* = \frac{P(u,u) - P(w,v) - \kappa_g + \kappa_f + \rho}{P(w,u) - P(w,v) + \rho}, \qquad P = A - LM. \tag{9}$$

Provided $P(w,u) - P(w,v) + \rho > 0$*, the club resists forgery for all* $\sigma < \sigma^*$*. When that denominator is negative the invasion advantage decreases in* σ *and the inequality reverses, so the club resists only above* σ^**; when it vanishes, which happens exactly when* $u = v$ *and* $\rho = 0$*, the forger's advantage is the constant* $P(w,u) - P(u,u) + \kappa_g - \kappa_f$*, independent of* σ *altogether.*

Equation (9) says something that is not obvious in advance: the club's in-group conduct enters the threshold twice, once through what members earn among themselves, $P(u,u)$, and once through what an impostor extracts from them, $P(w,u)$. A club whose members trust unconditionally maximises the second term, and so minimises its own tolerance for forgery. In a well-mixed population the reciprocating club (G, CS, CAS) survives up to $\sigma^* = 0.866$; the unconditionally trusting club (G, AS, CAS) survives only to 0.400. At the baseline assortment $r = 0.1$ the pair is 0.962 against 0.444, because a forger's self-encounters are worth much less to it than a member's are. The ratio is 2.17 at both, so it is the conduct and not the structure that sets it. Membership conduct is a security parameter.

The sign condition of Proposition 2 is not a technicality, and Section 3.8 is where it does its work: the denominator $P(w,u) - P(w,v) + \rho$ is positive for a club that treats outsiders harshly and non-positive for one that does not, so the two gentle out-group policies have no threshold to report rather than a low one. Figure 2 shows the invasion advantage as a function of σ in both structures, and the fixation probabilities that the finite-population process assigns to the same invasions.

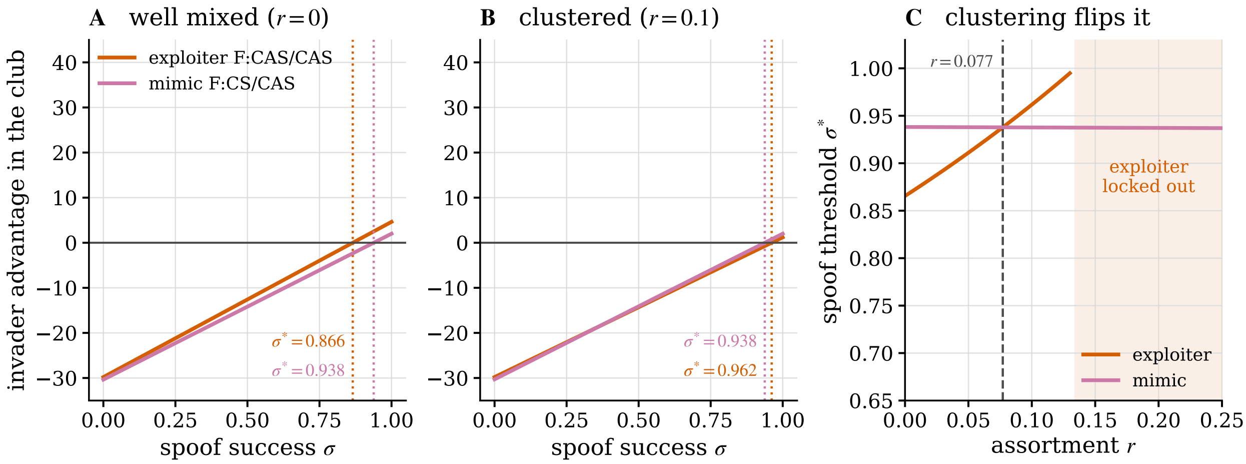

Figure 2: Clustering decides which of the two invaders reaches the club first. Advantage of a rare invader in a monomorphic club (G, CS, CAS), **(A)** well mixed and **(B)** at the baseline assortment $r = 0.1$; dotted verticals mark the thresholds, whose values are printed because a gap of 0.024 is not legible on a unit axis. **(C)** The same two thresholds against assortment. Clustering raises the exploiter's threshold steeply and leaves the mimic's flat, so the two cross at $r = 0.077$ and the exploiter is locked out entirely above $r = 0.134$. Better interaction structure removes the visible failure mode and not the invisible one.

3.3. Fines cannot buy what detection does not deliver

A natural response to forgery is to punish it. Setting the invasion advantage to zero and solving for the fine gives the smallest protective penalty.

Proposition 3 (Becker limit of attestation)**.** *In a well-mixed population, the smallest fine that keeps the unconditional forger out is*

$$\rho^*(\sigma) \;=\; \frac{\sigma\,[P(w,u) - P(w,v)] - [P(u,u) - P(w,v)] + \kappa_g - \kappa_f}{1-\sigma}, \tag{10}$$

which diverges as $\sigma \to 1$ whenever exploiting the club at full pass beats club membership. Where the club already resists unaided, that is for $\sigma < \sigma^*$, the expression is negative and no fine is required.

The expected penalty a forger faces is $(1-\sigma)\rho$, so as detection power vanishes the fine required to maintain a fixed expected penalty grows without bound. The divergence is the Becker trade-off between penalty size and detection probability [23, 72], and the numbers are stark: well mixed, $\rho^* = 11.9$ at $\sigma = 0.90$, 58.2 at 0.95 and 428.6 at 0.99, against a club payoff of 57.0. Beyond about $\sigma = 0.95$ the fine needed exceeds everything the club is worth. Assortment does not remove the pole, it only moves where it starts to bite: at the baseline $r = 0.1$ the club resists the exploiter unaided up to $\sigma^* = 0.962$, so nothing is required below that, and the requirement is then 8.6 at $\sigma = 0.97$ and 87.8 at $\sigma = 0.99$. What the model determines robustly is the divergence; where a given regime sits on the curve depends on its interaction structure.

The novelty is not the trade-off but which of its terms is available. In conventional enforcement the detection probability is bought with inspectors, so a regulator facing a high cost of detection can substitute a higher penalty and reach the same deterrence. Here the detection probability is a property of the attestation technology, and a regime that has deployed a weak one cannot buy its way out with penalties. The fine here is levied by an authority rather than by peers, so it escapes the second-order free-rider problem that constrains punishment in evolutionary settings where the punishing itself is costly and voluntary [73]; what constrains it instead is detection alone. Figure 3B shows the divergence.

Dues run the other way, and the closed form makes the direction unconditional. Differentiating Equation (9),

$$\frac{\partial \sigma^*}{\partial \kappa_g} \;=\; \frac{-1}{P(w,u) - P(w,v) + \rho} \;=\; -0.029 \tag{11}$$

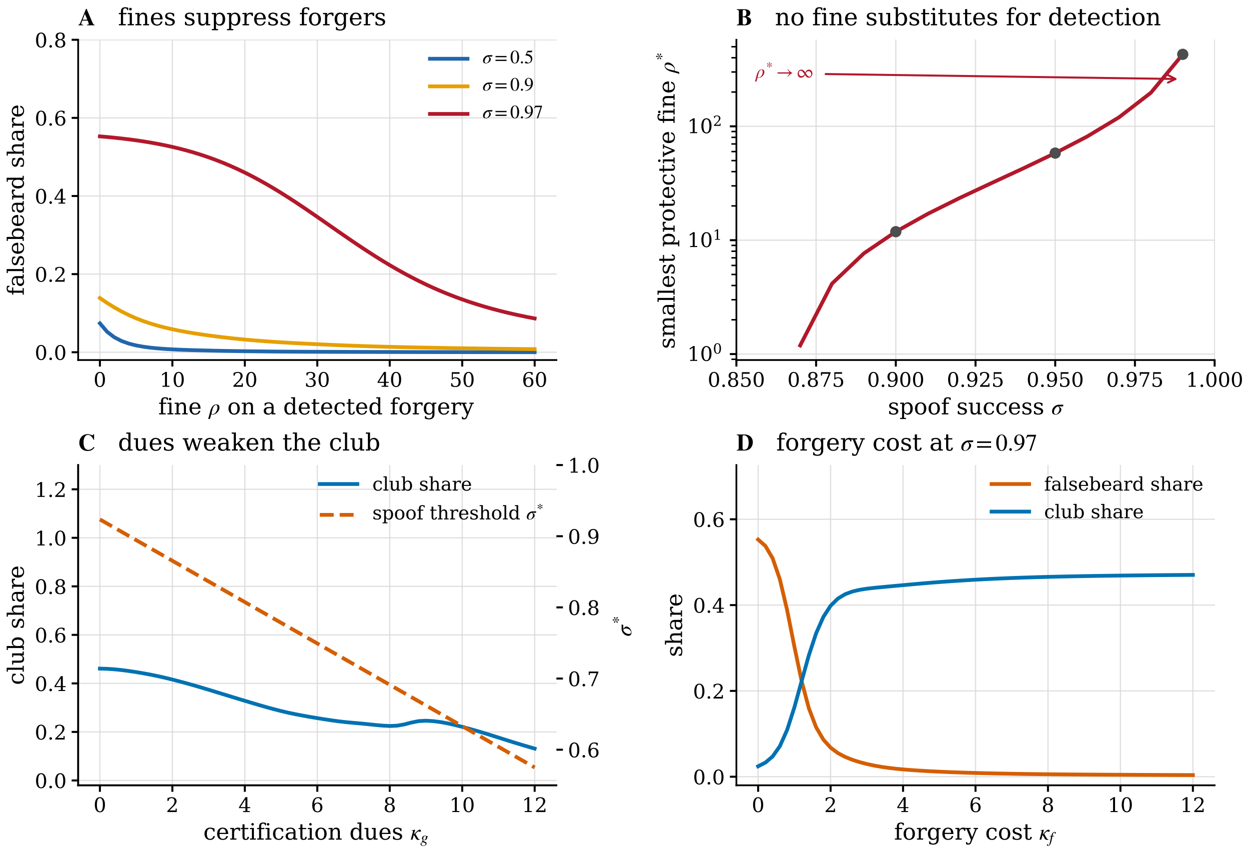

Figure 3: Four instruments, three of which have a limit. (A) Fines suppress forgers, and less so as detection weakens. **(B)** The smallest protective fine diverges as $\sigma \to 1$ (Proposition 3); note the logarithmic axis. **(C)** Certification dues weaken the club on both margins: the club share falls and the spoof threshold falls with it, by Equation (11). **(D)** Raising the cost of forgery is the one cost-side instrument that burdens the impostor rather than the member, shown at $\sigma = 0.97$ where the club would otherwise be extinct.

at the baseline, so every unit of certification cost lowers the club's forgery tolerance. Compliance cost burdens the compliant and never the impostor by construction, so every unit lowers forgery tolerance. This cost allocation is a modelling assumption, distinct from the information and intermediary-incentive problems studied in the certification literature [17]. The remedy is not cheaper dues but dearer forgery: raising κ_f moves σ^* up one for one, and Figure 3D shows it working at a spoof rate where nothing else does.

3.4. No badge starts a club by itself

Everything above concerns a club that already exists. Whether one can form is a separate question with a blunt answer.

Proposition 4 (Badge futility)**.** *Against any resident whose conduct does not condition on badges, a certified design earns exactly its unbadged twin's payoff minus the dues, at every σ, every ρ and every L. Consequently, at $r = 0$ and $\kappa_g > 0$, no certified design invades any unconditional resident that its unbadged twin could not already invade.*

The reason is that a badge is a signal, and a signal changes nothing when nobody reads it [24]. A club must therefore bootstrap from encounters with itself, which is what assortment supplies.

Proposition 5 (Nucleation threshold)**.** *Against an unconditional unbadged resident* (N, w, w)*, a club* (G, u, v) *invades exactly when*

$$r > r^* = \frac{P(w,w) - P(v,w) + \kappa_g}{P(u,u) - P(v,w)} = 0.063 \tag{12}$$

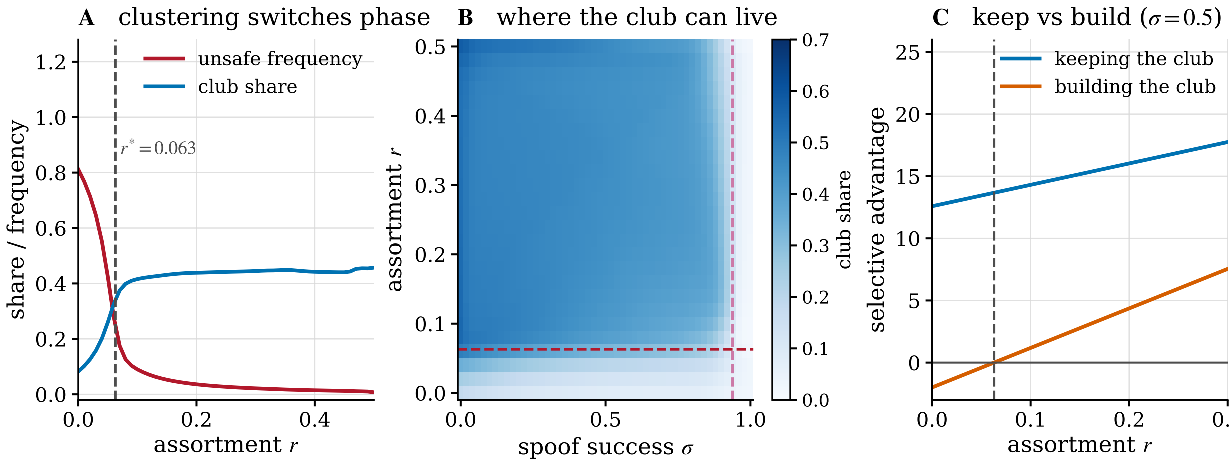

Figure 4: Clubs need clustering, and losing one is worse than never having had it. (A) Unsafe frequency and club share against assortment; the club switches on at $r^* = 0.063$ (Proposition 5). **(B)** Club share over spoof success and assortment. The horizontal line is r^* and the vertical line is the mimicry threshold σ_m; the club lives in the region between them. **(C)** The advantage of holding a club against forgery (blue) exceeds the advantage of building one from anarchy (orange) at every assortment, and the gap is what makes collapse irreversible.

at the baseline.

Taken together the two propositions say that verification quality and assortment are not substitutes. Attestation engineering, however good, moves σ, and σ does not appear in Equation (12). What does appear is the dues, which is why a certification regime that is expensive to join can be unable to start even when it would be stable once running. Figure 4 shows the phase structure: panel A the switch at r^*, panel B the two-dimensional region in which a club can live, and panel C the asymmetry between keeping a club and building one.

That asymmetry is the sixth result. At $r = 0$ a club cannot re-enter a population that has collapsed either to forgery or to anarchy, at any σ whatsoever: the re-entry threshold does not exist. The conditions for holding a certification regime together are strictly weaker than the conditions for assembling one, so a regime that is allowed to collapse is not recovered by restoring the policy that sustained it.

3.5. Collapse hollows out the mark before it degrades the conduct

Proposition 2 concerns a forger that exploits. A different invader is available, and it arrives first. The *mimic* (F, u, v) plays exactly the club's conduct and differs only in carrying a forged badge. It free-rides on the dues, not on the behaviour.

Proposition 6 (Mimicry threshold)**.** *The behavioural mimic invades the club when*

$$\sigma \;>\; \sigma_m \;=\; 1 - \frac{\kappa_g - \kappa_f}{P(u,u) - P(u,v) + \rho} \;=\; 0.938, \tag{13}$$

and at the baseline assortment $\sigma_m = 0.938$ *lies below the exploiter's threshold* 0.962.

The ordering of the two thresholds is not fixed, and what moves it is the one instrument that otherwise looks unambiguously good. Assortment penalises an exploiter, because an exploiter meets its own aggressive conduct in a self-encounter, and does not penalise a mimic, because a mimic meets the club's conduct, which is what a mimic plays. The exploiter's threshold therefore climbs from 0.866 at $r = 0$ to 1 by $r = 0.134$, beyond which no forgery quality whatsoever lets an exploiter in, while the mimic's threshold moves by 0.001 across the entire range. They cross at $r = 0.077$ (Figure 2C). At that point both transversal eigenvalues of Equation (6) at the club vertex vanish together, which makes $(r, \sigma) = (0.077, 0.938)$ a codimension-two organising centre for the identity layer: it is the parameter pair at which the ecosystem switches which of its two failure modes it meets first.

The consequence is uncomfortable for the instrument of Section 3.4. Clustering is what makes clubs possible at all, and it is also what removes the only failure mode that conduct monitoring can see. Past $r = 0.134$ the sole remaining

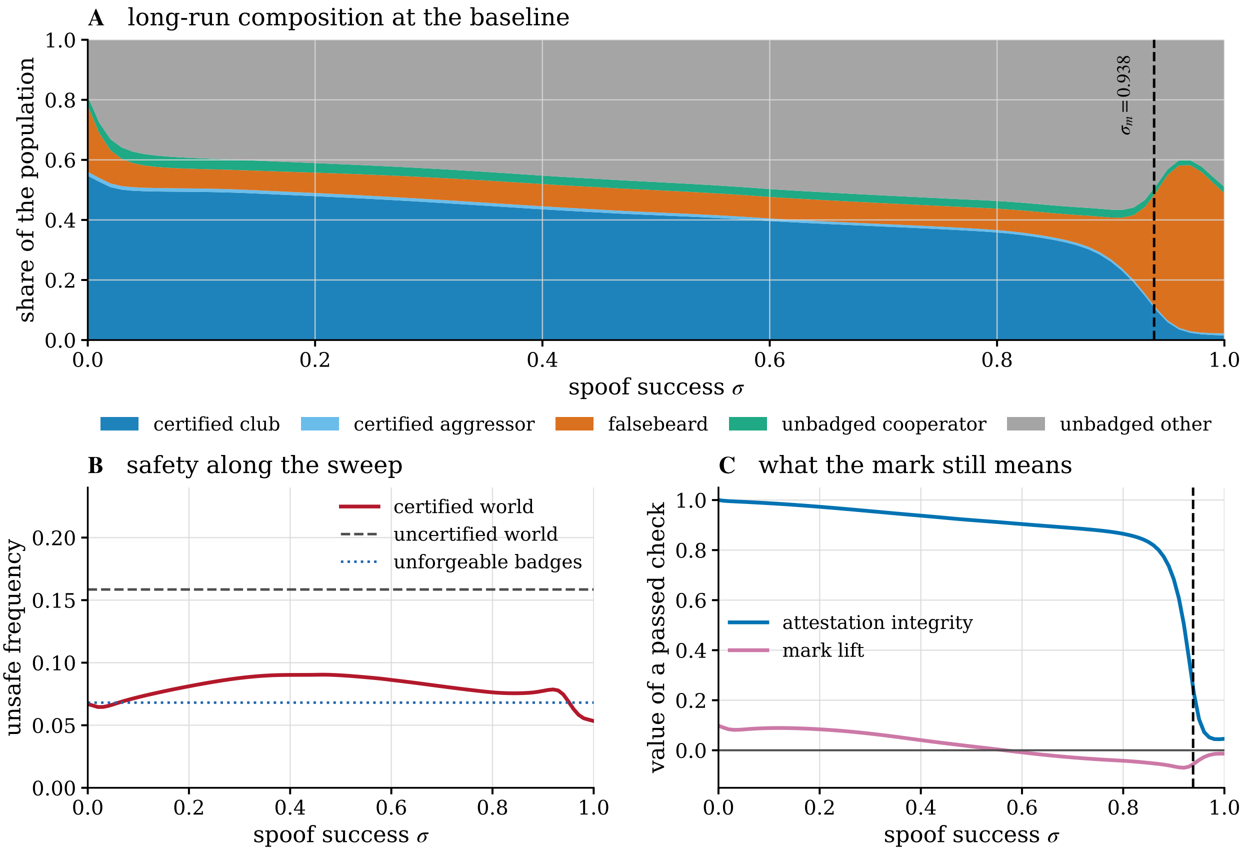

Figure 5: The certification cliff. (A) Long-run composition against spoof success; the dashed line is the mimicry threshold $\sigma_m = 0.938$. **(B)** Unsafe frequency of the certified world against the uncertified world and against a counterfactual in which badges cannot be forged. **(C)** The two things a passed check can be worth. Attestation integrity, the posterior that a passed badge is genuine, survives until σ_m; the mark lift, the behavioural value of a passed check, is already negative at $\sigma = 0.57$.

route into the club is impersonation by an agent that behaves correctly, so the better the interaction structure, the more completely the residual risk is concentrated in the failure that looks like success.

Between the two thresholds the ecosystem behaves well and means nothing. At $\sigma = 0.97$ forgers hold 55% of the population, the certified club has fallen to 0.02, the unsafe frequency is 0.058 against the baseline 0.090, and the posterior probability that a passed badge was actually issued has fallen to 0.05. An observer monitoring conduct would report that the certification regime is working, and would read the improvement in unsafe frequency as evidence for it. An observer asking what a certificate certifies would find that it certifies nothing.

Figure 5 traces the whole sweep. The composition in panel A shifts from the certified club to the falsebeard class as σ crosses σ_m. Panel C separates the two things a passed check could be worth: the attestation integrity falls through one half at $\sigma = 0.93$, while the mark lift, the behavioural value of a passed check, turns negative already at $\sigma = 0.57$. That inversion is worth stating plainly. Beyond $\sigma = 0.57$ a partner whose badge passes your check is *less* likely to treat you safely than one whose badge fails, because passing is what the exploiters do. The check has become an anti-signal [74]. We are careful not to overstate what follows. The certified world still has a lower unsafe frequency than the uncertified world at every σ (Figure 5B), so verification is not worse than nothing in aggregate. What is worse than nothing is the *advice* a passed check gives an individual agent, which above $\sigma = 0.57$ points towards the partners least deserving of trust. A protocol that instructs agents to extend credit on a passed credential is, in that regime, instructing them wrongly.

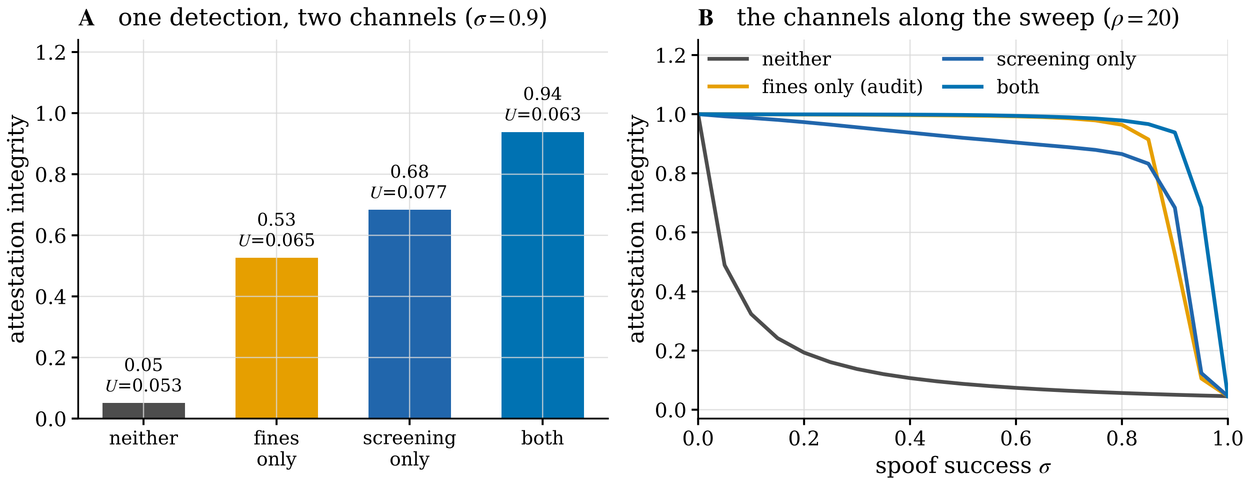

Figure 6: The two consequences of one detection. (A) Attestation integrity in the 2x2 at $\sigma = 0.9$, $\rho = 20$, with the unsafe frequency of each cell printed above the bar. Integrity varies by a factor of nearly twenty across cells in which the unsafe frequency barely moves. **(B)** The same four regimes along the spoof sweep.

3.6. Screening and fining are different instruments

One detection has two consequences, and governance regimes choose between them without always noticing. Real-time verification lets the counterparty act on a failed check; retrospective auditing levies a fine after the fact. The 2x2 of Figure 6 switches each off in turn at $\sigma = 0.9$ and $\rho = 20$.

The placebo cell, in which badges are checked but nothing follows from a failed check, leaves attestation integrity at 0.05: the population is essentially all forgers. Screening alone raises it to 0.68 and fines alone to 0.53; together they reach 0.94. The two channels are complements rather than substitutes, and neither reaches on its own what both reach together, which runs against the penalty-detection substitution of the standard enforcement model [23, 72].

The unsafe frequency tells a different story from the integrity, and this is the point of separating them. Across the four cells U moves only between 0.053 and 0.077, while integrity moves from 0.05 to 0.94. An evaluation regime that monitors conduct would rate all four cells as roughly equivalent and would be unable to distinguish a working attestation system from a completely hollow one. Attestation integrity has to be measured directly.

3.7. Provider-scoped marks make concentration a safety variable

Current agent-identity approaches use incompatible credential formats, trust roots and delegation semantics [75]. We model one plausible consequence: marks issued per provider need not be recognised by another provider. Consider K symmetric provider clubs, each running the club design with its own mark, with cross-provider checks failing by construction.

Proposition 7 (Concentration frontier)**.** *With symmetric shares the within-provider encounter probability equals the Herfindahl index* HHI $= 1/K$*, and the ecosystem unsafe frequency is*

$$U \;=\; \mathrm{HHI}\, U(u,u) \;+\; (1-\mathrm{HHI})\, U(v,v) \;=\; 1-\mathrm{HHI} \tag{14}$$

for the baseline club. Under federation, in which marks are mutually recognised [28], the ecosystem plays u throughout and $U = 0$ at any concentration.

A duopoly runs at $U = 0.500$ and eight providers at 0.875. Read naively this is an argument for concentration, which is not the reading we intend. It is an argument that scoping a safety mark to an issuer converts market structure into a safety variable, and that federation removes the dependence entirely.

The obstacle is that the incentive to federate runs backwards. Figure 7B shows the exclusion rent, the payoff gap between a club member and a compliant unbadged entrant that plays club conduct towards everyone and is treated as an outsider by every club. The rent is 30.4 under a single club and 2.5 across eight. A dominant provider therefore has the most to lose from mutual recognition [24], and a fragmented market has the least to gain from it while needing it

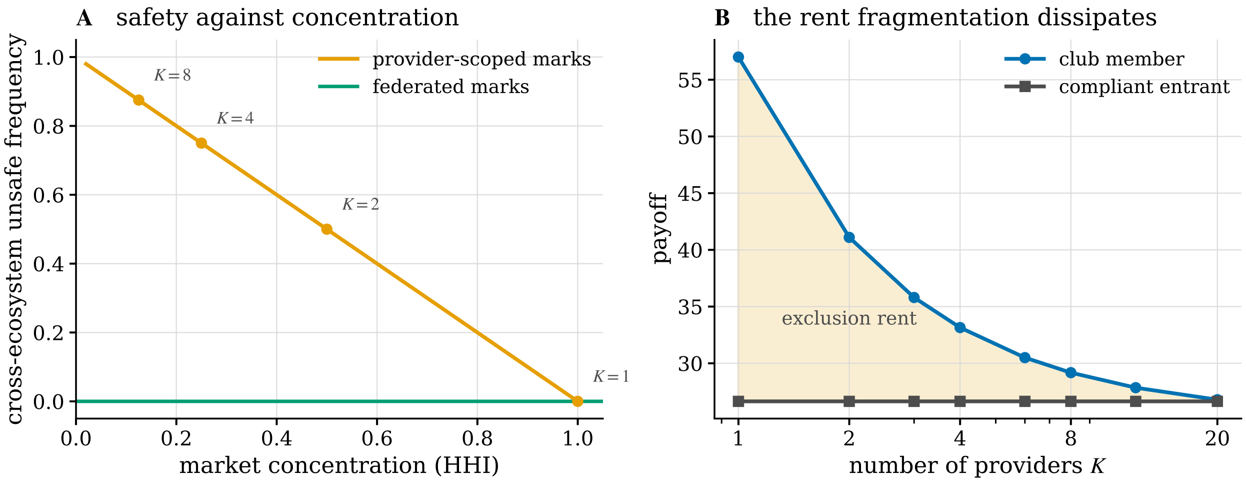

Figure 7: Scoped marks tie safety to market structure. (A) Cross-ecosystem unsafe frequency against concentration under provider-scoped marks is exactly $1 - \mathrm{HHI}$; federated marks are safe at any concentration. **(B)** The exclusion rent, the payoff gap between a club member and a compliant unbadged entrant, dissipates as the market fragments, so the provider best placed to federate is the one least willing to.

most. The exclusion rent is also a competition concern in its own right: in the model, a compliant entrant is penalised 30.4 under a monopoly mark for no reason connected to its conduct. This is the kind of entry barrier that equitable-interoperability proposals seek to reduce [29].

3.8. Bistability, and where the exclusion is actually paid

The results so far read as a qualified endorsement of certification: it works, it has thresholds, and the thresholds can be engineered. The eighth result qualifies the endorsement in a way that the thresholds do not reveal.

Everything above was computed in the small-mutation limit, which describes a market that is monomorphic almost always and moves between designs by mutation and fixation. The replicator flow asks a different question: from an arbitrary starting composition, what does the market settle into? The answer is not one thing.

Proposition 8 (Bistability)**.** *At the baseline the replicator flow has no badge-mixed attractor. All* 200 *interior starts converge to a rest point that is either entirely badged (*44% *of starts) or entirely unbadged (*56%*), with the replicator field vanishing to* 1.4×10^{-14} *at every one. Both regimes are internally safe: the largest unsafe frequency over all* 200 *attractors is* 1.5×10^{-22}.

The certified regime is dominated by the club (G, CS, CAS), which holds 0.85 of it, and the uncertified regime by the first-striker (N, CAS, CS), which holds 0.70. Both look unsafe by name and neither is: inside a certified regime every check passes and every member runs CS; inside an uncertified regime no check ever passes and every design runs its out-group conduct, which for the survivors is CS. Each population has selected a conduct that is safe against the only partner it ever meets.

The coincidence of the two safe regimes has an immediate and unflattering consequence for certification.

Corollary 9 (No safety dividend in a settled market)**.** *Since both regimes are safe, the social payoff of the certified regime differs from the uncertified one only by the certification cost:* 57.0 *against* 59.0*, a gap of* $2.0 = \kappa_g$ *to machine precision. In a market that has settled, the identity layer buys no safety and costs exactly its dues.*

The small-mutation result of Section 3 is not contradicted by this, and putting the two together is the point. Under rare mutation the certified world runs at 0.090 against the uncertified 0.159, so certification does reduce harm there. What it reduces is not the harm of a settled market, which is zero either way, but the harm the market suffers while being knocked out of safety and finding its way back. Certification buys *robustness*, not *level*. A regulator who measures a running, undisturbed ecosystem will find the certified and uncertified versions equally safe and the certified one poorer by its dues, and will be measuring the wrong thing.

Table 6
The out-group policy of a certified club decides both what it tolerates and what it costs an outsider. Each row restricts the design pool to one certified design plus every unbadged and forged design. The entry penalty is the payoff gap between a member and a compliant unbadged entrant that plays the club's own conduct towards everyone; the boundary unsafe frequency is what that encounter runs at. The two policies that impose no barrier are exactly the two that cannot survive forgery.

conduct towards outsiders	spoof threshold σ^*	entry penalty	boundary unsafe	club share (monomorphic)	certified basin
AS	0.000	−2.00	0.000	0.005	0.000
CS	0.000	−2.00	0.000	0.020	0.000
CAS	0.962	30.36	0.461	1.000	0.885
AU	0.962	40.40	0.868	1.000	0.395

Where the exclusion is. The club that survives is the one whose conduct towards outsiders is CAS, and the natural expectation is that the resulting harm shows up in encounters between the two regimes. Proposition 8 rules that out: in equilibrium the two regimes do not meet, so no such encounters occur. The exclusion is real all the same, and it is paid as a barrier rather than as a running harm.

Proposition 10 (The entry barrier)**.** *Let a compliant outsider be an unbadged design that plays the club's own in-group conduct towards everybody, so that it is behaviourally beyond reproach. Every club member's check of it fails, so every member executes its out-group conduct against it. Against the surviving club* (G, *CS*, *CAS*) *the outsider earns* 26.64 *where a member nets* 57.00*, a penalty of* 30.36 *for no conduct-related reason, and the encounter runs at unsafe frequency* 0.461*. Against a gentle club* (G, *CS*, *CS*) *the penalty is* −2.00*: the outsider is* better off *than the member, by exactly the dues, and the encounter is entirely safe.*

The two halves of Proposition 10 are the whole argument. Table 6 scans the four possible out-group policies and the ordering is exact. The two gentle policies, AS and CS, impose an entry penalty of −2.00 and a boundary unsafe frequency of 0; both have spoof threshold exactly 0 and equilibrium share 0.02 or less. The two harsh policies, CAS and AU, impose penalties of 30.36 and 40.40 and boundary unsafe frequencies of 0.461 and 0.868; both have spoof threshold 0.962 and take essentially the whole population under rare mutation. Nothing in between appears, because a club that is gentle to outsiders has made its badge behaviourally empty, at which point the dues buy nothing and any forger saves them.

There is an obvious objection, and answering it sharpens the result rather than softening it. If the problem is that a gentle club gives an impostor nothing to lose, then supply the loss externally: fine detected forgery, and the badge becomes worth acquiring honestly again. The first half of that works.

Corollary 11 (A fine restores the barrier, not the purpose)**.** *A fine on detected forgery raises the gentle club's spoof threshold from exactly* 0 *at* $\rho = 0$ *to* 0.923 *at* $\rho = 60$*, so forgery no longer defeats it. Its equilibrium share nonetheless never exceeds* 0.032 *at any fine level tested, in either reading of the dynamics.*

The reason is Proposition 4. A fine changes what a forger faces; it does not change what a badge earns against residents who do not read badges. A gentle certified club is behaviourally indistinguishable from the unbadged cooperator (N, CS, CS), which supplies the same conduct without the dues, so the club is dominated whatever the fine. Resistance to forgery and a reason to exist are separate requirements, and out-group harshness is the only conduct in this game that supplies both. What the fine does buy is worth stating, since the corollary is not an argument for doing nothing: it raises attestation integrity from 0.16 to 0.98 and lifts the social payoff of the full design space from 39.1 to 43.6. It makes the credential meaningful. It does not make the gentle club exist.

It is worth being precise about what carries the result, because a reader who knows the parochial-altruism literature will expect a different mechanism. Choi and Bowles [48] obtain a similar inseparability from group selection: hostility is favoured because it produces the intergroup conflicts in which in-group altruism pays. Nothing of that kind is present here. There are no groups that compete, no conflict whose outcome anyone wins, and no selection above the level of the individual design. What drives our version is excludability. The dues κ_g are a real cost, and a club that treats outsiders exactly as it treats members converts its badge into a label that predicts nothing about conduct, so the dues

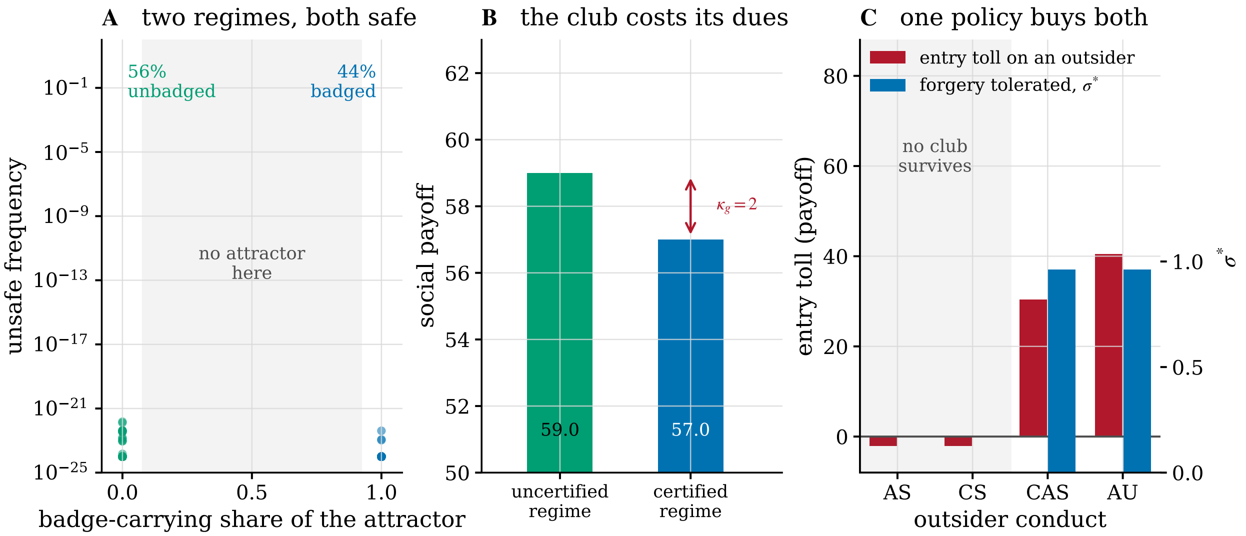

Figure 8: The market settles into one of two internally safe regimes, and the cost of certification is a barrier at the boundary rather than harm inside it. (A) Every one of the 200 interior starts of the replicator flow reaches a pure regime; none is badge-mixed, and the unsafe frequency of each is below 10^{-21}. **(B)** Social payoff of the two regimes. Both are safe, so the certified one is poorer by exactly the dues $\kappa_g = 2$, and the small-mutation reading, in which certification does reduce harm, measures robustness rather than level. **(C)** Across the four possible policies towards unverified partners, the forgery a club tolerates and the toll it charges a compliant outsider rise together. In the shaded region the club charges no toll, its spoof threshold is zero, and it does not survive.

purchase nothing and the cheapest way to carry the badge is to forge it. Equation (9) shows this directly: when $u = v$ the denominator $P(w, u) - P(w, v) + \rho$ collapses to ρ, so at $\rho = 0$ the forger's advantage does not depend on σ at all and is positive because the forger saves the dues. Out-group harshness is the only thing that makes the in-group benefit excludable. In this model it raises the potential cost of cheating relative to honest acquisition, the kind of off-equilibrium cost that can maintain honesty without a realised equilibrium signal cost [53].

A note on how not to compute this. An earlier version of this analysis averaged the replicator attractors into a single state and evaluated the observables there. Every observable in this model is bilinear in the population state, so scoring one at the mean of two disjoint attractors invents encounters between designs that never meet: the mean of the certified and uncertified faces assigns 44.4% of encounters to badged-unbadged pairs and reports an unsafe frequency of 0.379, when every attractor being averaged has zero such encounters and an unsafe frequency below 10^{-21}. The error is easy to make and hard to see, because the mean state is a perfectly valid probability vector. We report it here because the corrected reading is the more interesting one, and because a reader comparing against a multistable model of their own should know which of the two quantities they are looking at.

3.9. What identity is worth as liability varies

Proposition 1 predicts that identity should matter where liability cannot. Figure 9A measures it. The certification value, the unsafe frequency of the uncertified world minus that of the certified world at the same parameters, averages 0.054 below L_R and 0.002 above it, a factor of 22. Only 15% of the (σ, L) grid shows a value above one percentage point, and that region is concentrated at low liability and low spoof success.

The practical reading from our model is that attestation infrastructure is worth building where attribution is impossible. Agent-infrastructure work identifies traceability and accountability as central governance needs [4]; relevant settings include cross-border deployments, open-weight models with no traceable operator, and agent marketplaces with pseudonymous sellers. Where an operator can be identified and a liability regime applied [22], our model says the same money is better spent on that regime, which does not carry the exclusion cost of Section 3.8.

3.10. Robustness

Table 7 and Figure 9 collect the checks.

Table 7
Process robustness of the baseline equilibrium. Left: population size and selection intensity. Right: both readings of the setback clause of the race.

Z	β	unsafe	club	integrity
50	0.01	0.330	0.28	0.78
50	0.05	0.154	0.39	0.88
50	0.2	0.084	0.42	0.92
100	0.01	0.226	0.35	0.84
100	0.05	0.090	0.42	0.92
100	0.2	0.057	0.45	0.94
200	0.01	0.140	0.39	0.89
200	0.05	0.060	0.43	0.94
200	0.2	0.044	0.46	0.95
setback scope		unsafe	club	uncertified
total		0.090	0.42	0.159
prize		0.933	0.03	1.000

Process. Across population sizes 50, 100, 200 and selection intensities 0.01, 0.05, 0.2, the baseline unsafe frequency ranges from 0.044 to 0.330. The upper end is the weak-selection corner ($\beta = 0.01$), where the process is close to neutral drift and no design is strongly favoured; the ordering of the results does not change.

Two dynamics. The small-mutation and replicator readings agree on the location and direction of the attestation collapse, with a correlation of 0.965 between their integrity curves along the σ sweep, the replicator figure being the basin average of the integrity at each attractor. They differ on the level of the unsafe frequency, and Section 3.8 explains why: they are answering different questions about a bistable flow, not disagreeing about one answer.

Payoff reading. Under the alternative reading of the setback clause, in which a realised setback removes only the prize and not the accumulated stage payoffs, every threshold moves but the phase structure does not: L_R rises from 0.551 to 1.486, σ^* falls from 0.866 to 0.676, σ_m is essentially unchanged at 0.934, and r^* rises from 0.063 to 0.095. Because the baseline $r = 0.1$ then sits just above r^*, the club is marginal at the baseline under this reading and the certified world runs at 0.933; raising assortment to $r = 0.3$ restores the club and yields a certification value of 0.598, the largest in the study. The honest summary is that the thresholds are what the model determines robustly, and that where a particular baseline sits relative to them is reading-dependent.

Closed forms. Every closed-form threshold agrees with brute-force root-finding on the exact matrices to 2×10^{-16}, across dues, fines and liabilities.

Design pools. Removing forgery from the design space lowers the baseline unsafe frequency from 0.090 to 0.068, so forgery at a coin-flip spoof rate costs about a quarter of what unforgeable badges would deliver. Removing badges entirely raises it to 0.159.

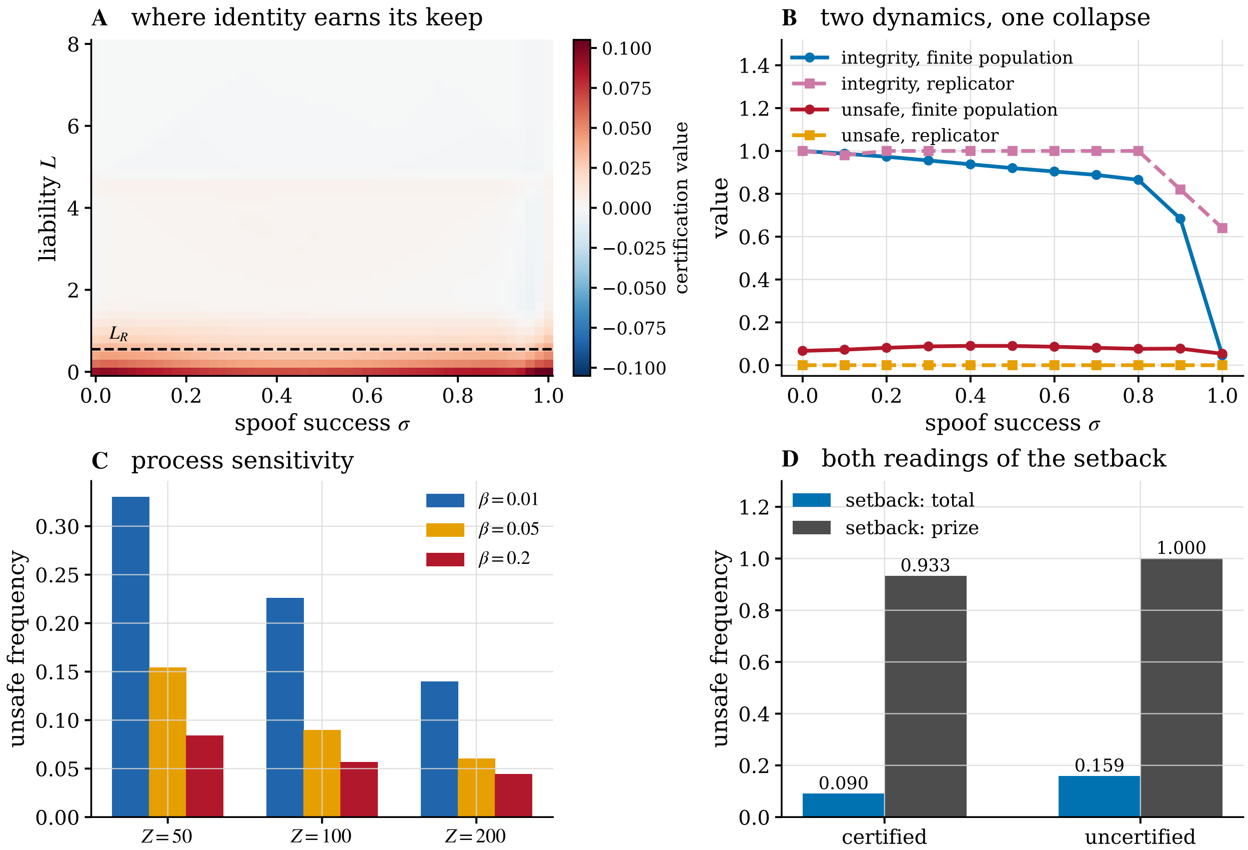

Figure 9: Robustness. (A) Certification value over spoof success and liability; the dashed line is L_R. The instrument earns its keep almost entirely below it. **(B)** The finite-population and replicator readings along the spoof sweep: they agree on the collapse of attestation integrity and disagree on the level of unsafe frequency, which is the finding of Section 3.8. **(C)** Population size and selection intensity. **(D)** Both readings of the setback clause.

4. Discussion

Three of our results bear directly on decisions being made now.

Attestation should be specified with its conduct rule, not without it. Standards work on agent identity [1, 25, 26, 75] concentrates on the verification problem: how to bind a credential to an agent, how to check it, how to revoke it. Our results say the verification problem is the easier half. Equation (9) shows that what a club does with a verified partner determines how much forgery the credential survives, and Proposition 10 shows that what it does with an unverified one determines the entire harm profile of the ecosystem. A credential specification that leaves conduct to the market has left the load-bearing part undefined [76].

Measure attestation integrity, not conduct. Section 3.6 shows a regime whose unsafe frequency is stable at about 0.06 across configurations whose attestation integrity ranges from 0.05 to 0.94. Conduct monitoring cannot distinguish a certification system that works from one that has been completely hollowed out by mimics, because a mimic behaves. The quantity to instrument is the posterior that a passed credential is genuine, which requires sampling and independently re-verifying credentials rather than observing outcomes. Section 3.5 makes the point sharper: the healthier the interaction structure, the larger the share of residual risk that conduct monitoring is blind to, because clustering removes exploitation and leaves impersonation untouched. A regime that has invested successfully in provider-scoped platforms has thereby made credential auditing more important, not less.

Federate early, because the incentive decays. The exclusion rent falls from 30.4 to 2.5 as the market fragments from one provider to eight, while the safety cost of non-federation rises from 0 to 0.875. The window in which a dominant provider can be induced to accept mutual recognition is therefore early, and the argument for it becomes strongest exactly when the party who must agree has least to gain.

On the exclusion result. We want to be careful about what Propositions 8 and 10 do and do not say. They do not say that certification is bad. Under rare mutation it halves the unsafe frequency, and below L_R it is the only instrument that works at all. They say that its benefit is robustness rather than level, and that its cost is a barrier rather than a harm [77]. A settled certified market is exactly as safe as a settled uncertified one and poorer by its dues, while a compliant outsider is charged 30.36 for conduct nobody objects to. Whether that trade is worth making depends on how often the market is knocked out of safety and on how many participants sit outside the club, neither of which the model can settle and both of which are measurable in a deployed system. What the model does settle is that the trade cannot be avoided by asking clubs to be gentle towards outsiders: that policy has spoof threshold zero, and no fine repairs it (Corollary 11).

On multistability. The methodological point generalises past this model. Multistable evolutionary dynamics are common [20, 30, 78], and so is the practice of summarising a population by a single averaged state [32], and summarising one by a basin-weighted mean state is a natural thing to do. Any observable that is bilinear in the state, which includes every population average of a pairwise quantity, must not be evaluated there: the mean of two pure regimes carries cross-terms that neither regime contains. In our case that error turned a safe ecosystem into one with an unsafe frequency of 0.379 and manufactured a boundary carrying 44.4% of all encounters where the true figure is zero.

4.1. Limitations

The interaction layer is a two-player reduced race with four designs, inherited deliberately and without modification, so that every effect we report is attributable to the identity layer rather than to a change in the underlying game, and it is not a model of any particular deployment. The reduction to four designs means the conduct space is coarse; a richer conduct space would allow intermediate out-group policies that our scan cannot represent, and the sharp zero-or-one structure of Table 6 would likely soften into a frontier. We expect the direction of the trade to survive, since it follows from Equation (9), but not the exact numbers.

Verification is a single Bernoulli draw with a fixed σ, identical across agents. A real attestation system has heterogeneous verifiers, revocation and reputation carried between encounters in the manner of indirect reciprocity [79]. It may also face adaptive forgers who improve their forgeries as detection improves, as demonstrated for watermark stealing [15]. The last of these is the most consequential omission: our σ is exogenous, and an arms race in which σ responds to enforcement would change the comparative statics of Section 3.3.

Assortment is modelled as self-encounter probability, the standard reduction, which conflates meeting one's own design with meeting one's own provider. A network model in which providers are communities of many designs would separate them, and the nucleation threshold would then depend on the community structure rather than on a scalar.

Finally, the two dynamics we report answer two questions and neither is a description of a running market. Actual agent ecosystems are neither monomorphic nor sitting at a replicator rest point, and they are not, in general, in either of the two pure regimes of Proposition 8. Which of our two readings is the better guide depends on how often the ecosystem is perturbed: the small-mutation reading governs a market that is repeatedly knocked off its equilibrium, and the replicator reading a market left alone. A model with an explicit perturbation rate would interpolate between them and would be the natural next step.

One consequence of the bistability is worth naming as a limitation rather than a result. Because no attractor mixes badged and unbadged designs, the boundary conduct we identify as the cost of certification is never exercised in equilibrium, and Proposition 10 measures it off the equilibrium path, as the payoff a compliant entrant would receive. Measuring it there is the right choice for an entry barrier, which is by definition what a party outside the market faces, but it does mean the model produces no prediction about the realised volume of cross-boundary harm. In a market where both regimes coexist for exogenous reasons, that volume would be the quantity to measure, and our model does not supply it.

5. Conclusion

We made a forgeable identity badge a strategic variable in an evolutionary AI race and characterised the resulting dynamics in closed form. Identity is the instrument of last resort: below the reciprocity threshold of the race no unbadged population holds safe play, and above it the identity layer is nearly worthless. A club's resistance to forgery is set by its in-group conduct, and conditional cooperation buys more than twice the tolerance of unconditional trust. Fines cannot substitute for detection, because the protective fine diverges as forgeries become undetectable, and in an agent economy detection is technological rather than budgetary. No badge starts a club without assortment, and none rebuilds one after collapse. Mimics hollow out the credential before exploiters degrade the conduct, so a monitoring regime that watches behaviour sees nothing wrong while the mark stops meaning anything.

The result that reframes the rest concerns what certification is worth once a market has settled. The replicator flow is bistable, with a certified and an uncertified regime and nothing in between, and both are internally safe. A settled certified market is therefore no safer than an uncertified one and poorer by exactly its dues; what certification buys is robustness, the protection of a safe market against being knocked out of safety, which is why it shows up under rare mutation and not at a rest point. Its cost is a barrier: a compliant outsider that behaves exactly as club members do is charged 30.36 for carrying no badge. That toll is not incidental. It is the only conduct under which the club survives forgery at all, since a club that is safe towards outsiders has spoof threshold zero and no fine repairs it. Certification and exclusion are the same mechanism, and a governance regime that adopts the first has adopted the second whether or not it says so. The instrument that escapes the trade is federation, which does not ask any club to be gentle towards strangers but removes the category of stranger instead.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

All code, generated tables and figures are openly available in the replication package at `https://github.com/trungkiet2005/greenbeard-tag`. Every scalar quoted in this manuscript is regenerated by `scripts/run_analysis.py` into `results/key_numbers.json` and checked against the text by `scripts/check_numbers.py`; every figure is rendered by `scripts/make_figures.py` and checked for layout collisions by the accompanying test suite. The interaction and identity layers carry no simulation: the race is evaluated exactly over the horizon law, and the handshake of Equation (1) is an exact four-term expectation. The one sampled quantity in the study is the basin distribution of the replicator flow, estimated from 200 initial conditions drawn uniformly from the simplex under a fixed seed; every attractor reached is a rest point to within 1.4×10^{-14}, and the quantity being estimated is the share of the simplex draining to each of the two regimes.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used a large language model assistant in order to draft and edit prose, to scaffold parts of the analysis code, and to check the consistency of quoted numerical values against the generated results files. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. No text, figure or numerical result was accepted without independent verification: every scalar in the manuscript is regenerated from the replication package by an automated check, and every analytical threshold is verified against brute-force root-finding on the exact payoff matrices.

CRediT authorship contribution statement

Trung-Kiet Huynh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review and editing. **Dao Sy Duy Minh:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review and editing. **Chi-Nguyen Tran:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review and editing.

A. Proofs

Proof of Proposition 1. Against a CS resident, the private payoff of a rare CAS mutant is $P(\mathsf{CAS},\mathsf{CS}) = A(\mathsf{CAS},\mathsf{CS}) - L\,M(\mathsf{CAS},\mathsf{CS})$ and the resident earns $P(\mathsf{CS},\mathsf{CS})$. The difference is affine in L with slope $-[M(\mathsf{CAS},\mathsf{CS}) - M(\mathsf{CS},\mathsf{CS})] < 0$, so it is positive exactly below the ratio in Equation (7). For the general statement, a safe unbadged resident plays a safe design in self-play; since badges are absent, every check fails and only s_{out} is ever executed, so the resident is behaviourally one of $(\mathsf{N},\cdot,\mathsf{AS})$ or $(\mathsf{N},\cdot,\mathsf{CS})$. With assortment r, the mutant meets itself with probability r, so Equation (6) gives its advantage over a $(\mathsf{N},\cdot,\mathsf{CS})$ resident at $L = 0$ as $r\,A(\mathsf{CAS},\mathsf{CAS}) + (1-r)\,A(\mathsf{CAS},\mathsf{CS}) - A(\mathsf{CS},\mathsf{CS})$, which is affine and decreasing in r and vanishes at Equation (8). The same calculation against a $(\mathsf{N},\cdot,\mathsf{AS})$ resident gives $42.77 - 74.57\,r$, which vanishes only at $r = 0.574$, so the $(\mathsf{N},\cdot,\mathsf{CS})$ residents are the binding ones and $r^{\dagger}$ is the smaller of the two roots. The invasion of $(\mathsf{N},\mathsf{CAS},\mathsf{CAS})$ against each of the eight such residents is verified directly on the exact matrices at $r = 0$, and the four $(\mathsf{N},\cdot,\mathsf{CS})$ residents are verified to resist at the baseline $r = 0.1$. □

Proof of Proposition 2. In a monomorphic club, a rare unconditional forger (F,w,w) meets club members. Its badge passes with probability σ, in which case the member plays u; otherwise the member plays v. The forger plays w regardless. Its payoff is therefore $\sigma P(w,u) + (1-\sigma)P(w,v) - \kappa_f - (1-\sigma)\rho$ and the resident earns $P(u,u) - \kappa_g$. Both are affine in σ; equating and solving gives Equation (9). The denominator $P(w,u) - P(w,v) + \rho$ is positive whenever exploiting a trusting member beats being shut out, which holds for every (u,v,w) used here. □

Proof of Proposition 3. Solve the same indifference condition for ρ instead of σ. The numerator of Equation (10) is the forger's advantage at zero fine, and the denominator $1-\sigma$ is the probability that a fine is levied. If the numerator is positive at $\sigma = 1$, that is if $P(w,u) - \kappa_f > P(u,u) - \kappa_g$, then $\rho^* \to \infty$. □

Proof of Proposition 4. Let the resident be (N,w,w). Its conduct does not depend on the focal badge, so it plays w whatever the focal design carries. The focal design's own check of the resident always fails, since the resident carries no badge, so the focal plays s_{out}. The encounter is therefore (s_{out},w) for both a certified and an unbadged focal design with the same conduct pair, and the payoffs differ only by κ_g. Since σ enters only through the pass rate of a forged badge, which is not present, the identity holds at every σ; the same argument gives independence of ρ and L enters both sides identically. The conclusion follows because at $r = 0$ a rare mutant meets only residents. □

Proof of Proposition 5. With assortment r, a rare club mutant meets itself with probability r and the resident otherwise. Its fitness is $r[P(u,u) - \kappa_g] + (1-r)[P(v,w) - \kappa_g]$, since in a self-encounter both badges pass and both play u, while against the unbadged resident the club plays v and the resident plays w. The resident earns $P(w,w)$. Rearranging the inequality gives Equation (12). □

Proof of Proposition 6. The mimic (F,u,v) in a club population plays u when its badge passes and v when it does not, and the member reciprocally plays u or v. Its expected payoff is $\sigma P(u,u) + (1-\sigma)P(u,v) - \kappa_f - (1-\sigma)\rho$ against the resident's $P(u,u) - \kappa_g$. Solving for the indifference point gives Equation (13). The mimic's threshold lies below the exploiter's whenever the exploiter's conduct is worse in self-play than the club's, which is what makes the exploiter pay a larger assortment penalty. □

Proof of Proposition 7. Under symmetric shares, an agent of provider k meets an agent of the same provider with probability $1/K$, which equals $\mathrm{HHI} = \sum_k (1/K)^2 \cdot K$. Within-provider checks pass and both play u; cross-provider checks fail and both play v. Equation (14) is the resulting mixture. For the baseline club, $U(\mathsf{CS},\mathsf{CS}) = 0$ and $U(\mathsf{CAS},\mathsf{CAS}) = 1$, giving $U = 1 - \mathrm{HHI}$. Under federation every check passes and the mixture collapses to $U(u,u)$. □

Proof of Proposition 10. The decomposition is an identity, not an approximation. Let w_{ij} be the probability of the ordered pair (i, j) under the encounter law and let $\mathcal{B}$ partition the ordered pairs by the badge status of the two seats. Then $U = \sum_{ij} w_{ij}u(i, j) = \sum_{B\in\mathcal{B}} \sum_{(i,j)\in B} w_{ij}u(i, j)$, so the block contributions sum to U exactly. The numerical values are evaluated at the replicator attractor. The zero entries are exact in the model: a club member facing a club member executes $(\mathsf{CS}, \mathsf{CS})$, which contains no unsafe action, and the surviving unbadged designs execute $(\mathsf{CS}, \mathsf{CS})$ against each other for the same reason. $\square$

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